

Transparent, Post-Quantum Anonymous Set Membership from FRI-STARKs over BabyBear and Mersenne-31

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*A construction-and-analysis paper: two security obligations (O1, O4) are stated as explicitly open;
the remaining goals are established here.*

Scope. A self-contained description of a transparent anonymous set-membership argument and its security analysis, presented as a generic primitive. The building blocks (FRI-STARKs, Poseidon, sponge hashing, nullifier-based membership) are standard; the contribution is the *concrete instantiation and its parameters* together with an explicit security analysis. Two obligations are stated as **explicitly open** and analyzed rather than proved: the $\text{GF}(p^2)$ -state Poseidon round counts (O1, specific to Arm A) and the formal zero-knowledge simulator’s masking-degree budget (O4, shared by both arms). The remaining goals (membership soundness and anonymity, Theorems 1–2, together with the sponge obligations O2, O3 and the proof-system soundness of Section 6.5) are established here (Section 6), each conditional on the obligations it names. Leaving O1 and O4 open while proving the reductions around them is deliberate: it isolates exactly what independent cryptanalysis must settle.

Abstract

We describe and analyze a transparent, plausibly post-quantum argument of *anonymous set membership with a nullifier*: a prover demonstrates, in zero knowledge, that it knows a secret t whose leaf $L = H(t)$ lies in a public Merkle accumulator with root $root$, and simultaneously publishes a context-bound nullifier $N = H(domain \parallel t \parallel ctx)$ that is unlinkable across contexts but stable within one. The argument is a FRI-STARK over a small 31-bit prime field, using a single algebraic hash (a Poseidon-family permutation in a wide-sponge mode) for the leaf, every Merkle node, and the nullifier. Transparency (no trusted setup) and a hash-only soundness assumption make the construction conservative against quantum adversaries.

We give a complete security analysis along four axes: (O1) algebraic / Gröbner-basis security of the permutation round counts; (O2) collision and second-preimage resistance of the truncated-output sponge; (O3) domain separation between the leaf and nullifier pre-images; and (O4) the zero-knowledge simulator for the hiding-PCS instantiation. We **implement, measure, and compare** two field instantiations of the hash. Arm A runs the permutation *state* over the quadratic extension $\text{GF}(p^2)$ of Mersenne-31; Arm B runs over the base field of BabyBear with the deployed Poseidon2. Both are implemented, and their proof-system soundness is computed to 128-bit post-quantum (Section 6.5); at equal security, **Arm B’s proofs are $\approx 14\text{--}18\times$ smaller and verify $\approx 16\text{--}17\times$ faster**, and we recommend it. We prove membership soundness and anonymity as reductions (Theorems 1–2) conditional on two obligations we leave explicitly open and analyzed rather than proved: the extension-field round-count parameterization *specific to Arm A* (O1), and the zero-knowledge simulator’s masking-degree budget (O4, shared by both arms).

Keywords: anonymous set membership · nullifiers · zero-knowledge proofs · FRI-STARKs · transparent setup · post-quantum cryptography · Poseidon / Poseidon2 · algebraic hashing · BabyBear / Mersenne-31.

1 Introduction

Anonymous set membership with nullifiers underlies anonymous signaling, one-person-one-vote / rate-limiting, private credentials, and privacy-preserving messaging (e.g. Semaphore-style signaling [Sem], rate-limiting nullifiers [RLN], and privacy pools). The standard shape is: members are accumulated in a Merkle tree over a commitment $H(t)$ of a secret t ; to act, a member proves in zero knowledge that its leaf is in the tree and reveals a deterministic nullifier N derived from t and a context, so that double-action *within* a context is detectable while actions *across* contexts are unlinkable.

Most deployed instantiations use pairing- or discrete-log-based SNARKs (Groth16, PLONK) with a trusted or universal setup, and rely on assumptions broken by a quantum adversary. We target a **transparent** (setup-free) and **plausibly post-quantum** instantiation by realizing the relation as an AIR proven with a **FRI-STARK** [BBHR18, BGKS20], whose soundness rests only on collision resistance of a hash and the FRI low-degree test. The single algebraic hash is a Poseidon-family permutation [GKRRS21, GKS23] in a wide-sponge mode, chosen for low AIR degree.

A second post-quantum route is *lattice-based* succinct ZK, e.g. LaBRADOR [LaBRADOR] and the recent toolkit of Biasioli et al. [BBL+26] that adds zero-knowledge to it, which yields markedly smaller proofs (under 100 KB) but rests on structured-lattice assumptions (Module-SIS / LWE). We deliberately take the more conservative *hash-only* path (its sole assumption is collision resistance and it needs no trusted setup), and target a purpose-built membership relation rather than a general prover (Section 5).

Contributions.

1. A self-contained description of the membership AIR: trace layout, in-circuit sponge constraints, Merkle path threading, the nullifier sub-circuit, and the public-input binding (Sections 3–4).
2. Two concrete, **implemented and measured** field instantiations of the hash: a $\text{GF}(p^2)$ -state variant over Mersenne-31 (Arm A) and a base-field Poseidon2 variant over BabyBear (Arm B), with their digest widths, security margins, wire sizes, and a side-by-side cost comparison (Section 5).
3. A four-axis security analysis (O1–O4, Section 6) carried out against the published round-count and sponge-indifferentiability bounds, including an adversarial self-check against the recent algebraic-attack literature.
4. A precise statement of the two residual questions on which we seek external review (Section 7).

We make no claim that the underlying techniques are new. The value we hope to add is a *published, reviewable* concrete parameter set and analysis for a transparent, hash-only anonymous-membership argument over a small prime field.

2 Preliminaries

Field. Arm A uses $p = 2^{31} - 1$ (Mersenne-31), a 31-bit prime; $\text{GF}(p^2)$ denotes the quadratic extension realized as $\text{Complex}\langle\text{GF}(p)\rangle$ with $i^2 = -1$ (valid since -1 is a non-residue: $p \equiv 3 \pmod{4}$);

a $\text{GF}(p^2)$ element packs two base limbs, $\log_2 |\text{GF}(p^2)| \approx 62$. Arm B (Section 5.2) instead uses the **BabyBear** prime $q = 2^{31} - 2^{27} + 1$ (also 31-bit, but two-adic of order 2^{27}) as its base field. Field-generic statements below are written over F .

Algebraic permutation. We use a Poseidon-family permutation π over F^t (state width t) with an x^α S-box: $\alpha = 5$ for Arm A ($\gcd(5, p-1) = \gcd(5, p^2-1) = 1$, so x^5 permutes both $\text{GF}(p)$ and $\text{GF}(p^2)$); $\alpha = 7$ for Arm B (since $5 \mid q-1$ makes x^5 non-bijective over BabyBear). There are R_F full rounds and R_P partial rounds, ARC (add-round-constant), and an MDS / Poseidon2 external+internal linear layer. Round constants are derived by an XOF / Grain-LFSR from a fixed domain string.

Sponge. H is the sponge [BDPV07] over π with rate r , capacity c ($t = r + c$), 10*1 padding, and a *truncated* output equal to the first w_{out} cells of the final state. We use H for the leaf $L = H(t)$, every Merkle node $H(\text{left} \parallel \text{right})$, and the nullifier $N = H(\text{domain} \parallel t \parallel \text{ctx})$.

FRI-STARK / AIR. The relation is expressed as an Algebraic Intermediate Representation (AIR): an execution trace (a matrix over the field) and a set of polynomial constraints of bounded degree, proven low-degree after the relation is reduced to a quotient via FRI [BBHR18]. We use a STARK whose trace, constraints, and FRI all live over a common field F (Section 5 discusses $F = \text{GF}(p^2)$ for Arm A vs. the two-adic base-field BabyBear configuration for Arm B; both run on a radix-2 FRI domain, not a circle-STARK [HLP24]).

Zero-knowledge. Plain STARK traces are not zero-knowledge. We obtain ZK via a *hiding* polynomial commitment: the trace is low-degree-extended at an increased blow-up with vanishing-polynomial randomization, Merkle commitments are salted with fresh CSPRNG output, and FRI runs on the randomized codeword [Hab22, BCRSVW19]. Only the public statement ($\text{root}, \text{ctx}, N$) is revealed.

Threat model and security goals

Adversary. We consider a probabilistic polynomial-time adversary $\mathcal{A}$; the concrete parameters of Section 6.5 additionally target a quantum $\mathcal{A}$, through the hash-only soundness assumption and the post-quantum challenge-field sizing. $\mathcal{A}$ sees the circuit, its baked constants, the public statement ($\text{root}, \text{ctx}, N$), and every proof; it may choose the accumulator contents, insert its own leaves, and adaptively obtain honest members' proofs. The accumulator root is supplied by the verifier and trusted; the domain constant is derived from a fixed public generator (Section 3).

Goals. We phrase three goals as games. Section 6 discharges those that do not depend on the open obligations O1 / O4, and states the two principal ones as conditional theorems.

Definition 1 (Membership soundness). Let $\mathcal{P}^*$ produce an accepting $(\text{root}, \text{ctx}, N, \pi)$ and let $\mathcal{E}$ be the proof system's knowledge extractor. $\mathcal{A}$ wins if the verifier accepts yet $\mathcal{E}$ extracts no (t, path) with $\text{MerkleVerify}(\text{root}, H(t), \text{path}) = \text{true}$ and $N = H(\text{domain} \parallel t \parallel \text{ctx})$. Its advantage is $\text{Adv}_{\mathcal{A}}^{\text{snd}}$.

Definition 2 (Anonymity within the set). $\mathcal{A}$ names two members i_0, i_1 of the accumulator and a fresh context ctx^* ; the challenger proves membership for i_b , $b \xleftarrow{\$} \{0, 1\}$, and returns (π, N) . Its advantage is $\text{Adv}_{\mathcal{A}}^{\text{anon}} = |\Pr[\mathcal{A} = b] - \frac{1}{2}|$.

Definition 3 (Nullifier stability and cross-context unlinkability). Within a fixed ctx a member’s N is deterministic, so double-action is detectable. Across distinct contexts the family $\{N_{ctx}\}$ of one member is required to be computationally indistinguishable from independent uniform values, and no $\mathcal{A}$ lacking that member’s t can produce an accepting proof carrying its N .

3 The Membership Relation

Fix a domain constant $domain$ (a baked circuit constant, *not* a witness or public input), derived as the first $|domain|$ cells of $H(S)$ for a fixed statement string S . The argument proves, for public $(root, ctx, N)$:

$$\begin{aligned} \exists (t, path) : & \text{MerkleVerify}(root, L = H(t), path) = \text{true} \\ & \wedge N = H(domain \parallel t \parallel ctx) \\ & \text{reveal only } (root, ctx, N); \quad L \text{ and } t \text{ remain secret.} \end{aligned}$$

$root$ is the verifier-supplied, trusted accumulator root; ctx is a public per-context label; N is the nullifier. Membership soundness reduces to: a prover cannot make the AIR accept unless it folds a genuine $H(t)$ to the trusted $root$ and emits the correctly-derived N . Anonymity reduces to: the proof, $root$, and N reveal nothing about which leaf L was used (within the anonymity set of the tree) and N is unlinkable across distinct ctx .

4 Construction (the AIR)

Trace. One row per Merkle level, plus row-0 carrying the leaf and nullifier sponge blocks. The public values are the w_{out} -cell $root$, the $|ctx|$ -cell ctx , and the w_{out} -cell N .

In-circuit sponge. Each hash invocation is unrolled into $\lfloor L/r \rfloor + 1$ permutation applications (L = input length in cells). Every round intermediate is pinned by exactly one constraint: ARC, the x^α S-box, and the linear layer in full rounds; in partial rounds the non-active lanes are copy-constrained so they cannot be freely chosen. Capacity is threaded as the linear-layer *expression* between permutations (no free capacity column), and the digest is the first w_{out} cells of the genuine final-state expression, never a free read-back column. The absorb + 10*1 padding schedule matches a value-level reference, checked across input lengths.

Merkle threading. Per level: a boolean dir selects the node input order ($(running, sibling)$ vs. $(sibling, running)$); $running$ is updated to the node hash; a transition constraint forces $running' = parent$ (next-row $running$); the last row binds $parent = root$. Trees of varying depth are handled by deterministically extending $root$ with zero-sibling levels (padding-blind in the AIR; soundness rests on the verifier feeding its *own* trusted $root$).

Nullifier. The $|domain|$ -cell $domain$ prefix is injected as constant expressions (no committed domain column). The pre-image is exactly $domain \parallel t \parallel ctx$. The single committed t slice feeds *both* the leaf sponge and the nullifier pre-image (so leaf and nullifier are structurally same- t). Row-0 gated constraints bind $running = L$, ctx to the public ctx , and the nullifier output to the public N .

Degree. Max constraint degree is α (the ungated x^α S-box: 5 for Arm A, 7 for Arm B); selectors gate only degree-1 bindings. The quotient-chunk count is derived from the constraint AST by both prover and verifier, so any degree overflow fails at prove time.

The constraint set. Collecting the above, the AIR enforces five constraint families, each pinned as described:

- C1. Leaf sponge.** Row-0 evaluates $L = H(t)$ over the committed t (ARC, x^α , linear layer, and 10×1 padding).
- C2. Merkle threading.** Each level applies $H(\text{left} \parallel \text{right})$ in the *dir*-selected input order, updates $\text{running}' = \text{parent}$, and the last row binds $\text{parent} = \text{root}$.
- C3. Nullifier preimage.** The same committed t , prefixed by the constant *domain* and suffixed by *ctx*, evaluates $N = H(\text{domain} \parallel t \parallel \text{ctx})$.
- C4. Public binding.** $\text{running} = L$ at row 0, *ctx* equals the public context, and the nullifier output equals the public N .
- C5. Degree / FRI.** The ungated maximum constraint degree is α , and the quotient-chunk count is derived from the constraint AST, so any degree overflow fails at prove time.

A structural audit of this AIR (sponge under-constraint, Merkle threading, leaf / same- t , nullifier integrity, padding rows, padding-root, public binding, degree / FRI config) found no relation a malicious prover can exploit to accept a proof that violates the decoded statement, *modulo* the cryptographic assumptions O1–O4 below.

5 Field Instantiation

The single design lever with security and cost consequences is the **field over which the hash permutation state runs**. We describe, implement, and measure both arms. **Both are realized, and their proof-system soundness computes to 128-bit post-quantum (Section 6.5); Arm B is the instantiation we recommend.** Arm A runs the permutation state over the quadratic extension $\text{GF}(p^2)$ of Mersenne-31; Arm B runs over the base field of BabyBear with the deployed Poseidon2. Measured at equal security, Arm B’s proofs are $\approx 14\text{--}18\times$ smaller and verify $\approx 16\text{--}17\times$ faster (Section 5.4), and its round-count obligation (O1) is a near-citation of a standard parameter set rather than the novel extension-field analysis Arm A requires. All other aspects of the construction (Sections 3–4) are identical between the two.

5.1 Arm A: $\text{GF}(p^2)$ -state (the analyzed instantiation)

State width $t = 7$, rate $r = 2$, capacity $c = 5$, $R_F = 8$, $R_P = 60$, $\alpha = 5$, over $\text{GF}(p^2)$. Each state cell is a $\text{GF}(p^2)$ element (≈ 62 bits). Digest $w_{\text{out}} = 5$ cells $= 5 \times 8 = 40$ bytes $= 5 \times 62 \approx 310$ bits. The STARK trace, constraints, and FRI all run over $\text{GF}(p^2)$.

- **Pro:** the wide ≈ 62 -bit cells give a 5-cell, 310-bit digest and 155-bit capacity collision resistance with only $c = 5$.
- **Con:** running the *whole* trace and FRI over $\text{GF}(p^2)$ pays a $\approx 2\times$ field tax on all prover arithmetic and committed data; and the published round-count formulas (Section 6, O1) target the *base* prime field, so the extension-field state requires the dedicated analysis of O1.

The FRI challenges are drawn from a **degree-3 extension** $\text{GF}(p^6) \approx 186$ **bits**: the ≈ 62 -bit value field is too small to serve as its own challenge field at 128-bit soundness, so the membership prover / verifier lift the challenge / OOD sampling into $\text{GF}(p^6)$ (Section 6.5).

Public statement: $\text{root}(40) \parallel \text{ctx}(16) \parallel N(40) = 96$ bytes; per-cell encoding 4-byte real $\parallel$ 4-byte imaginary, canonical.

5.2 Arm B: base-field BabyBear / Poseidon2 (the realized, recommended instantiation)

State width $t = 16$, $R_F = 8(4+4)$, $R_P = 13$, $\alpha = 7$, over the **BabyBear** prime field $q = 2^{31} - 2^{27} + 1$ ($\log_2 q \approx 30.9$), using the *deployed* Poseidon2 parameters [GKS23, Plonky3]: the external block-circulant 4×4 MDS layer, the optimized internal diagonal, and Grain-LFSR round constants from the reference generator, i.e. the width-16 instance shipped by Plonky3 / SP1. BabyBear, not Mersenne-31, is the base field for a deliberate reason: M31’s base field has 2-adicity 1 (no radix-2 FFT subgroup), so a base-M31 STARK would require *circle* STARKs [HLP24]; BabyBear’s 2-adicity 27 gives a native two-adic FFT domain, so Arm B runs on the **same TwoAdicFriPcs** machinery as Arm A; only the field and permutation change, not the proof system. BabyBear’s S-box is x^7 ($\alpha = 7$, since $5 \mid q - 1$ makes x^5 non-bijective).

To preserve a ≥ 256 -bit digest with ≈ 31 -bit cells, the capacity and output widen to $\lceil 256/30.9 \rceil = 9$ cells: sponge $t = 16$, $r = 7$, $c = 9$, $w_{\text{out}} = 9$ (≈ 278 -bit capacity $\rightarrow \approx 139$ -bit collision; $9 \times 4 = 36$ -byte digest). FRI challenges are drawn from a degree-5 extension $\text{GF}(q^5) \approx 155$ bits, sized for 128-bit soundness (Section 6.5).

- **Pro:** O1 reduces from a novel extension-field analysis to a near-*citation* of the standard, widely-deployed BabyBear Poseidon2 parameter set (Plonky3 / SP1); and the base-field trace (4-byte cells) avoids the $\text{GF}(p^2)$ field tax across the whole proof. Measured, the effect is decisive (Section 5.4).
- **Realization:** Arm B is fully implemented and tested **for functional correctness**: the field, the deployed Poseidon2 (KAT against the published reference vector), the in-circuit gadget, the wide sponge / Merkle, the membership AIR, and a transparent **and** zero-knowledge prover / verifier, with an exhaustive per-column under-constraint audit and a negative-proof matrix over all sub-AIRs. This is functional evidence, **not** a soundness proof: like Arm A, Arm B remains open on O1 / O4 (Sections 6–7). Public statement: $\text{root}(36) \parallel \text{ctx}(16) \parallel N(36) = 88$ bytes, per-cell 4-byte canonical little-endian.

5.3 What the field choice does and does not change

The field choice governs O1 (round-count security) and, through it, O2 / O3 (collision and domain-separation, which assume a secure permutation). It does **not** affect O4 (the zero-knowledge simulator), which is a property of the hiding PCS and the masking-degree budget, not of the hash field. A reviewer’s verdict on O4 transfers unchanged between the two arms.

5.4 Measured cost (both arms, equal 128-bit-PQ security)

Both arms are implemented and benchmarked at their 128-bit-PQ configurations (Arm A: $\text{GF}(p^6)$ challenge, `log_blowup` 3; Arm B: $\text{GF}(q^5)$ challenge, `log_blowup` 4; both bind on a SHAKE256 Merkle commitment at 128 bits; Section 6.5). Single-thread, median of 5, membership depth 32:

	Arm A ($\text{GF}(p^2)$ / Poseidon256)	Arm B (BabyBear / Poseidon2)	Arm B advantage
Trace row width	17 152	1 661	$\approx 10\times$ narrower
Bytes per trace row	137 216	6 644	$\approx 21\times$ smaller
Proof size, transparent	≈ 16.4 MB	≈ 1.04 MB	$\approx 16.6\times$ smaller
Proof size, zero-knowledge	≈ 16.7 MB	≈ 1.21 MB	$\approx 14.4\times$ smaller
Verify time	≈ 510 ms	≈ 30 ms	$\approx 17\times$ faster

The gap is **at equal security** (both reach the same 128-bit hash floor), not a size-for-security trade. It tracks bytes-per-trace-row: FRI opens $\approx (\#queries) \times (\text{trace width})$ field elements, and Arm A’s Poseidon256 ($t = 7$, 68 rounds, 8-byte $\text{GF}(p^2)$ cells) yields a far wider, heavier row than Arm B’s Poseidon2 ($t = 16$, 21 rounds, 4-byte cells). This field / permutation effect, not the FRI blowup, dominates, so it holds across depths and in both transparent and zero-knowledge modes.

6 Security Analysis (O1–O4)

The structural audit shows the constraints faithfully encode the intended hash and Merkle relations. The following are the cryptographic obligations that an automated / structural argument cannot discharge. We give engineering derivations against the published formulas; we do **not** claim these are a substitute for independent expert review (Section 7).

What is proved here

The structural audit (Section 4) and the proof-system characterization (Section 6.5) let us state the two principal goals as conditional theorems. Each is *conditional on the open obligations it names*: we prove the reductions, not the obligations. The obligations O1–O4 themselves are then analyzed in the remainder of this section.

Theorem 1 (Membership soundness). *Assume O1 (so, by O2, the sponge H is collision-resistant at the 128-bit level) and the knowledge-soundness of the FRI-STARK (Section 6.5). Then for any probabilistic polynomial-time $\mathcal{P}^*$,*

$$\text{Adv}_{\mathcal{P}^*}^{\text{snd}} \leq \varepsilon_{\text{ks}} + \varepsilon_{\text{coll}},$$

where ε_{ks} is the knowledge-soundness error of the proof system and $\varepsilon_{\text{coll}}$ the sponge-collision advantage.

Proof sketch. Knowledge-soundness extracts a full satisfying trace from any accepting $\mathcal{P}^*$, except with probability ε_{ks} . By the per-round, per-column audit of Section 4, a satisfying trace encodes a genuine sponge evaluation $L = H(t)$ (C1), a Merkle path whose every node is a genuine $H(\text{left} \parallel \text{right})$ folding to the last-row *parent* bound to the trusted *root* (C2), and a nullifier output equal to $H(\text{domain} \parallel t \parallel \text{ctx})$ over the *same* committed t (C3–C4). An accepting trace that violated the decoded statement would therefore exhibit two distinct preimages sharing a node or digest, i.e. a sponge collision, which occurs with probability at most $\varepsilon_{\text{coll}}$ under O2 (itself resting on O1). Summing the two failure events gives the bound. $\square$

Theorem 2 (Anonymity and unlinkability). *Assume O4 (the prover is zero-knowledge under the hiding PCS), O3 (leaf and nullifier preimages are domain-separated), and that H behaves as a pseudorandom function of ctx for a fixed secret t (the standard random-permutation modeling of the sponge, consistent with the PRP premise of O1). Then $\text{Adv}_{\mathcal{A}}^{\text{anon}}$ and the cross-context linking advantage are negligible.*

Proof sketch. By O4 a simulator produces π from $(\text{root}, \text{ctx}, N)$ alone, so π reveals nothing about which leaf was used beyond the fact of membership; anonymity within the set is thus the anonymity of the accumulator itself. The only other public value is $N = H(\text{domain} \parallel t \parallel \text{ctx})$, which by O3 is domain-separated from the leaf; under distinct ctx the sponge inputs differ, so by the pseudorandomness assumption the nullifiers are indistinguishable from independent uniform values, giving cross-context unlinkability, while stability within one ctx is immediate from the determinism of H . $\square$

The obligations these theorems rest on are analyzed next.

O1: Round-count / algebraic-degree security

Instance (Arm A). $\alpha = 5$, $t = 7$, $R_F = 8$, $R_P = 60$ (68 rounds) over $\text{GF}(p^2)$, $\log_2 |F| \approx 62$. Target $M = 128$ -bit.

We evaluate the published Poseidon round-count bounds [GKRRS21] (§5.5), [ABM23]: statistical, interpolation, and the Gröbner-basis family, reproduced from the canonical generators. With $n = \log_2 p$ the field-size term, $\log_5 2 \approx 0.43068$:

- **Statistical:** $R_F \geq 6$ if $M \leq \lfloor n - (\alpha - 1)/2 \rfloor (t + 1)$, else $R_F \geq 10$.
- **Interpolation:** $R_F + R_P \geq 1 + \lceil \log_5 2 \cdot \min(M, n) \rceil + \lceil \log_5 t \rceil$.
- **Gröbner:** several bounds, incl. $(t - 1)(R_F + R_P) \geq (t - 2) + M / (2 \log_2 \alpha)$ and the Macaulay-style $\text{cost}_{\text{gb}} \geq M$.
- **Margin:** +2 full rounds and +7.5% partial rounds over the minimum.

Regime	Binding bound (interpolation)	Margined optimal (R_F, R_P)	Deployed 8 + 60
$n = 62$ (realistic $\text{GF}(p^2)$)	$1 + \lceil 0.43068 \cdot 62 \rceil + 2 = 30$	(8, 26), total 34	$\approx 2 \times \text{total} / 2.3 \times R_P$
$n = 31$ (pessimistic subfield)	$1 + \lceil 0.43068 \cdot 31 \rceil + 2 = 17$	(8, 14), total 22	$\approx 4.3 \times R_P$

cost_{gb} for (8, 60) evaluates to ≈ 349 bits $\gg 128$. So 8 + 60 clears every published bound under both the realistic and worst-case-subfield substitutions.

Extension-field subtlety (the crux for review). $\text{GF}(p^2)$'s tower structure could *reduce* security only via a **subfield descent**: if $\text{GF}(p)^7$ were invariant under the round function, an attacker could work in the 31-bit subfield ($\min(M, n) \rightarrow 31$) and unlock invariant-subspace / subspace-trail Gröbner attacks [Bariant+22, GKR25]. The defense is **generic round constants**: drawing both the real and imaginary limb of every constant independently from the XOF gives each constant a pseudorandom nonzero imaginary part ($\Pr[\text{imag} = 0] \approx 2^{-31}$ per constant), so a single off-subfield constant per round maps $\text{GF}(p)^7$ off itself and breaks subfield invariance. *Caveat for the reviewer*: if the linear (MDS) layer is built from base-field entries, the linear layer alone preserves $\text{GF}(p)^7$; the subfield defense then rests **entirely** on the constants' genericity. A future change that de-randomized the constants' imaginary parts would re-expose the subfield.

Adversarial self-check. The 2024–2026 algebraic-attack wave (the FreeLunch / CheapLunch Gröbner attacks [FreeLunch, CheapLunch] and subspace-trail Gröbner cryptanalysis of Poseidon [GKR25]) pressures R_P by linearizing partial rounds; $R_P = 60$ is 2.3–4.3 $\times$ the requirement, and the instances those attacks render borderline run R_P at $\approx 1 \times$. The off-envelope concern reduces to the constant-genericity check. Notably, [GKR25] finds that the *original* Poseidon round-count model can itself mis-estimate the rounds required for some instantiations (under- or over-counting); this makes the large $R_P = 60$ margin reassuring but is a further reason expert review, not formula evaluation alone, is warranted. The case does not survive automated scrutiny, but the residual against *future* cryptanalysis of a $\text{GF}(p^2)$ -state permutation (an unusual deployment choice) is real, and is why we seek review.

Arm B contrast. Arm B's permutation is the deployed BabyBear Poseidon2 [GKS23, Plonky3]: $\alpha = 7$, $t = 16$, $R_F = 8$, $R_P = 13$ over the base prime field ($\log_2 q \approx 30.9$), the **standard, widely-deployed** instantiation (Plonky3 / SP1) the round-count formulas target directly. O1 becomes confirmation that these published parameters are used unchanged (a base-field citation with no tower

/ subfield-invariance / off-envelope-state concern) rather than the novel extension-field analysis Arm A requires.

O2: Collision / second-preimage of the truncated sponge

For a sponge on a random permutation, collision and (second-)preimage resistance are, up to the capacity [BDPV08],

$$\min(2^{c \cdot \log_2 |F|/2}, 2^{w_{\text{out}} \cdot \log_2 |F|/2}).$$

- **Arm A:** capacity $5 \cdot 62 = 310$ bits $\rightarrow 2^{155}$; output $5 \cdot 62 = 310$ bits $\rightarrow 2^{155}$ and ≥ 256 -bit digest. Required capacity for $M = 128$ is $c \geq 2M/\log_2 |F| = 256/62 \approx 4.13$; $c = 5$ clears it. Truncation to $w_{\text{out}} = 5$ (≥ 4.13) preserves ≥ 128 -bit output collision resistance; *internal* collisions (which enable Merkle-node equivocation) are governed by the capacity, not the output width, so truncation does not weaken node-collision resistance. Sponges with nonzero capacity are not length-extendable.
- **Arm B:** capacity $9 \cdot 30.9 \approx 278$ bits $\rightarrow \approx 2^{139}$; 9-cell ≈ 278 -bit digest ≥ 256 bits (BabyBear ≈ 31 -bit cells; $c = 9 \geq 2M/\log_2 q \approx 8.3$ clears the 128-bit floor).

O2 holds *given* O1 (the permutation is a secure PRP at the 128-bit level).

O3: Domain separation (no leaf $\leftrightarrow$ nullifier cross-collision)

The leaf preimage is the secret t ; the nullifier preimage is the strictly longer *domain* $\parallel t \parallel ctx$; both use the same sponge with 10*1 padding. (i) *Cross-protocol*: a distinct *domain* constant forces a distinct absorption from round 1, giving disjoint nullifier images (standard prefix domain separation). (ii) *No leaf $\leftrightarrow$ nullifier impersonation*: the two preimages differ in length (so they absorb in a different number of permutations), and the nullifier carries the *domain* prefix the leaf lacks; a collision between them is therefore a generic cross-length collision (O2), not a structural ambiguity, because 10*1 padding is injective on field-sequence inputs. O3 adds only the discipline that the domain string is globally unique to this statement.

O4: Zero-knowledge simulator (field-independent)

Implemented. A hiding PCS (salted hiding-Merkle MMCS + FRI on a vanishing-poly-randomized codeword at $\log_blowup + 1$) keeps the witness (t, L , siblings, direction bits, all sponge intermediates) in the blinded trace; only $(root, ctx, N)$ is public. Two blinding-randomness leaks that we found and fixed: (L1) the salts / blinding polynomials must come from a CSPRNG, not a linear PRNG (a salted-but-linear generator is de-blindable when salts appear in the proof); (L2) the seeds must be fresh, independent, ≥ 256 -bit per proof, drawn from an OS CSPRNG, not fixed / guessable constants.

The residual we seek review on. A formal simulator $\mathcal{S}$ that, given only $(root, ctx, N)$, outputs proofs computationally indistinguishable from honest ones. The structure: $\mathcal{S}$ commits to a random masking polynomial of the same degree budget and answers FRI / OOD queries via the standard ZK-STARK simulation [BCRSVW19] + hiding-PCS zero-knowledge [Hab22]; indistinguishability reduces to (i) hiding of the salted Merkle commitments and (ii) the masking degree **strictly exceeding** the number of query + OOD openings, so every revealed evaluation is uniform and independent of the witness. The **quantitative residual**: with the degree- α S-box forcing $\log_blowup \geq 3$ and a concrete $\#queries + \#OOD$, the masking degree under the hiding PCS must be confirmed $\geq \#openings + 1$ for the production parameters, and the truncated-output

sponge openings (w_{out} of t cells) must leak nothing beyond the public digest under that budget. O4 is field-independent; it transfers unchanged between Arm A and Arm B.

6.5 Proof-system (PCS / FRI) soundness: the challenge field

O1–O4 concern the *hash*; the proof system has its own soundness, bounded by the FRI / DEEP challenge field and the query phase and binding on the Merkle commitment. The field-dependent terms (out-of-domain / DEEP / proximity-gap) are **capped by the size of the challenge field**, regardless of query count, a subtlety that cost us a real error: Arm A originally drew its challenges from its own ≈ 62 -bit value field $\text{GF}(p^2)$, which caps DEEP soundness near 62 bits, and a query-only estimate masked it. The fix is a larger challenge extension. Both arms now compute to 128-bit post-quantum soundness, **binding on a 256-bit SHAKE256 Merkle commitment** (128-bit collision, NIST Category 2):

- **Arm A:** challenge field $\text{GF}(p^6)$ (degree-3 over $\text{GF}(p^2)$, ≈ 186 bits), `log_blowup` 3, 96 queries, 20 PoW bits $\rightarrow$ DEEP / Schwartz–Zippel ≈ 176 bits; binding term 128 (SHAKE256).
- **Arm B:** challenge field $\text{GF}(q^5)$ (degree-5 over BabyBear, ≈ 155 bits), `log_blowup` 4, 96 queries, 20 PoW bits $\rightarrow$ DEEP / Schwartz–Zippel ≈ 147 bits; binding term 128 (SHAKE256).

The post-quantum model is the deployed QROM one: Fiat–Shamir preserves a round-by-round-sound IOP up to small factors, Grover halves only the grinding proof-of-work, and the SHAKE256 binding is the 128-bit term. This is a parameter characterization of the PCS layer; it assumes the AIR is sound and complete (the structural audit and per-column mutation tests are evidence, not proof) and that O1 holds.

7 The Open Obligations

Theorems 1–2 reduce the security goals to the four obligations; two of these, O1 and O4, are analyzed above but not proved, and independent cryptanalysis is decisive for each:

1. **(O1) The $\text{GF}(p^2)$ -state parameterization (Arm A).** Running the *entire permutation state* over $\text{GF}(p^2)$ (rather than the base field, with the extension reserved for FRI challenges) is an unusual deployment choice. We have shown $8 + 60$ clears the published bounds by $\approx 2\times$ and that generic round constants close the subfield-descent path, but we seek confirmation of: (a) the constant-genericity argument as the intended subfield defense; (b) the absence of a residual invariant subspace under the base-field MDS + generic-constant composition; (c) comfort with the $\text{GF}(p^2)$ -state convention against current and foreseeable algebraic cryptanalysis. *If a reviewer is uncomfortable, Arm B, the recommended, implemented base-field instantiation, avoids this obligation entirely, reducing O1 to conformance with published BabyBear Poseidon2 parameters.*
2. **(O4) The simulator’s masking-degree budget.** We seek a formal simulator write-up and confirmation that the concrete masking degree $\geq \# \text{queries} + \# \text{OOD} + 1$ for the production parameters, and that the truncated-output sponge openings leak nothing under that budget.

On **Arm A vs. Arm B:** we recommend Arm B on the measured merits (at equal 128-bit security its proofs are ≈ 14 – $18\times$ smaller and verify ≈ 16 – $17\times$ faster, Section 5.4) *and* on the cleaner O1 (a base-field parameter citation, not an extension-field analysis). Arm A is the original $\text{GF}(p^2)$

instantiation; we retain its full analysis because the $\text{GF}(p^2)$ -state construction is of independent interest, but the off-envelope O1 above is why we do not lead with it.

A reviewer wishing to reproduce the O1 evaluation needs only the parameters in Section 5 and the public generators cited below; no proprietary material is involved.

8 Related Work

- **Algebraic hashes.** Poseidon [GKRRS21] and Poseidon2 [GKS23] establish the round-count formulas and the base-field parameterizations we cite; the HADES design strategy and its algebraic cryptanalysis [ABM23], with the recent attack literature [Bariant+22, FreeLunch, CheapLunch, GKR25], inform the O1 self-check.
- **Sponge security.** The indifferenciability and truncated-output bounds are from Bertoni–Daemen–Peeters–Van Assche [BDPV07, BDPV08].
- **Transparent STARKs.** FRI and STARKs [BBHR18, BGKS20]; circle STARKs over Mersenne-31 [HLP24]; zero-knowledge STARKs / hiding PCS [BCRSVW19, Hab22]; the Plonky3 toolkit [Plonky3] provides the BabyBear Poseidon2 reference parameters (Arm B) and the two-adic FRI machinery both arms use.
- **Post-quantum proof systems (the other family).** The principal alternative post-quantum ZK route is *lattice-based*: LaBRADOR [LaBRADOR] and the succinct-ZK toolkit of Biasioli et al. [BBL+26] (building on it) give sub-100 KB proofs for general statements (the smallest among post-quantum schemes) at the cost of structured-lattice assumptions. We instead rest only on hash collision resistance (transparent, no algebraic structure), trading proof size for a more conservative assumption and a purpose-built membership relation.
- **Anonymous membership / nullifiers.** Semaphore-style signaling [Sem], rate-limiting nullifiers [RLN], the PLUME deterministic-nullifier scheme [PLUME], and privacy-pool constructions establish the leaf / nullifier pattern; our contribution is a transparent, hash-only (post-quantum-oriented) instantiation over a small prime field, with the accompanying parameter analysis.

Artifact Availability

All code, parameters, in-circuit audits, and the benchmarking harness behind the claims in this note are open source, so that the construction (Sections 3–4), the parameter sets and security analysis (Sections 5–6), and the measured comparison (Section 5.4) can be independently rebuilt and verified.

- **Implementation.** *Both arms* are implemented in the `lib-q-zkp` crate of the `libQ` library: <https://github.com/Enkom-Tech/libQ> (Apache-2.0), pinned at commit 34cb3f0 (tag v0.0.8). The shared in-circuit gadgets are `src/air/wide_sponge.rs`, `wide_merkle_path.rs`, and `wide_hash.rs`; the membership API is `src/membership.rs` / `src/circuit.rs` / `src/api.rs`. **Arm A** ($\text{GF}(p^2)$ / Mersenne-31): membership AIR `src/air/unlinkable_membership.rs`, gadget `src/air/poseidon_gadget.rs` ($\alpha = 5$), STARK config `src/stark.rs`, Merkle `src/merkle.rs`. **Arm B** (BabyBear / Poseidon2): membership AIR `src/air/unlinkable_membership_baby_bear.rs`, gadget `src/air/poseidon2_gadget.rs` ($\alpha = 7$), STARK config `src/stark_baby_bear.rs`, Merkle `src/merkle_baby_bear.rs`. Both expose transparent and zero-knowledge prover/verifier paths.

- **Reproduction package.** A companion repository bundles this paper’s source together with a pinned snapshot (git submodule) of `libQ` and a `REPRODUCE.md` that maps each claim to the file, test, or benchmark that backs it: <https://github.com/En-Cipher/transparent-pq-anonymously-membership>.
- **Security analysis (O1–O4).** The round-count / soundness derivations are recorded in `docs/membership-arm-a-soundness-params.md`, `membership-arm-b-soundness-params.md`, `membership-arm-b-obligation-packet.md`, `membership-adr113-freeze-gate-review.md`, and the adversarial self-check in `membership-arm-b-redteam.md`. The structural / under-constraint audit is exercised by `tests/air_integration.rs`, `tests/ip_soundness_tests.rs`, `tests/unlinkable_membership_tests.rs`, `tests/zero_knowledge_tests.rs`, and `tests/security_parameters.rs`.
- **Measured cost (Section 5.4).** Produced by `benches/zkp_benchmarks.rs` (harness `measure_arm_a` / Arm B), documented in `docs/membership-arm-b-measurement.md`. Environment: AMD Ryzen 9 5900X (12C/24T), Windows 10 IoT x86_64, Rust 1.96 (edition 2024), `cargo test --release`, single-threaded (the `parallel/rayon` feature off), `prove` and `verify` reported as the median of 5, proof size via `postcard::to_allocvec(proof).len()`. Absolute timings are hardware-dependent; the cross-arm *ratios* are the reproducible result.

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